Alibi Shokputov

Teaching Statement

My goal as a teacher is to help students become confident enough to investigate an unfamiliar problem and thoughtful enough to question their first answer. I am drawn to teaching analytical work because it makes reasoning visible: a choice about a measure or a spatial boundary can change how we understand a problem. I ask students to explain those choices and return to the question their analysis is meant to answer. A successful project, for me, combines sound technical work with an account of why the approach makes sense and where its conclusions stop.

In teaching Maryland's graduate Research Methods course, URSP601, I used a semester-long empirical study to connect research design with analysis in R. That structure gave me opportunities to discuss how an early choice could affect everything that followed. At project milestones, I could respond to the reasoning behind an analysis while students still had time to revise it. I see that feedback as a conversation about the student's question and the evidence available to answer it. When assessing a project, I care about how clearly a student explains a methodological decision and recognizes its consequences. A well-explained limitation can show me as much understanding as a correctly executed technique.

The materials I prepare are meant to make unfamiliar work approachable without hiding the decisions it involves. In URSP601, I provided datasets and R Markdown files connecting demonstrations with written interpretation. I wanted students to have a reference they could return to as they worked through a problem themselves. In a geospatial course, I would use a worked example to establish the reasoning behind an operation, then change the data or geographic scale. Asking students to adapt the analysis would help me see whether they could transfer the idea. I want to see how they respond when the example no longer supplies every answer and where they need support.

For geospatial teaching, I find questions about everyday urban conditions a useful starting point. Estimating jobs within thirty minutes, for example, requires decisions about travel mode and departure time. Comparing neighborhood conditions raises questions about boundaries and denominators. I would use these decisions to give spatial scale and data models a purpose before introducing more complex operations. Students could sketch an expected result, test it in GIS, and investigate differences between the two. I would stay close to their reasoning through this process, asking how a join changed the record count or what missing observations meant for the people represented.

My applied research keeps me attentive to what an analysis means for the people it represents. Studying environmental conditions around housing, for example, requires me to consider how site boundaries and spatial resolution affect a comparison. I would bring questions like these into class through public data or prepared examples. Working with public agencies has also made me care about the audience for an analysis. I would ask students to explain how someone might use their findings and what that person would need to know before relying on them. For me, communicating a limitation is part of doing useful analytical work.

I welcome the different ways students from other disciplines understand a problem. I would invite them to explain what their field helps them notice and which assumptions they would question. A student who is new to programming may bring substantive knowledge that changes how a team interprets its results. I want that contribution to be visible while also helping the student develop technical confidence. Early exercises would help me identify where support was needed, while extensions would give experienced students room to explore. In group projects, I would rotate technical and interpretive responsibilities so each student had opportunities to develop both. Individual explanations would help me understand where someone was ready for greater independence or needed further guidance.

Student feedback has made me more attentive to balancing explanation and practice. In URSP601, students valued project milestones and timely feedback but asked for more supervised coding time. I adjusted

assignments when the workload competed with completing the final study. That experience reinforced my interest in seeing students work through a problem while I can respond to their questions. Questions between meetings would guide what I revisit. In online courses, I would pair recorded demonstrations and annotated examples with project checkpoints and scheduled help sessions, giving students regular opportunities to discuss work in progress.

I want student projects to have a possible life beyond the final report. In capstone courses and graduate advising, I would discuss potential audiences early enough to shape a feasible project. Interested master's and doctoral students could work toward a conference submission or publication when the question and evidence warrant it. If they wanted to develop that work further with me, I would be open to coauthoring a publication. Other projects might produce maps or analyses for community partners. A wider audience can motivate students and strengthen their portfolios. I would help them choose feasible next steps that serve their goals.

My own learning shapes how I think about teaching. When experimenting with tools such as Claude Code and NotebookLM, I am interested in what they let me attempt and how I can judge whether their assistance is useful. Developing cross-platform workflows, including agent-assisted programming, keeps me engaged with unfamiliar approaches. It also makes verification an active part of learning. I would bring that curiosity into evaluating geospatial AI assistants for instruction and be willing to revise an exercise as I learned what helped students understand the work.

In class, I would explain permitted AI uses and ask students to define their analytical task. An assistant might help interpret an error or suggest a workflow; I would then ask students how they checked its output against documentation and data. Disclosure of assistance and brief individual explanations would help me assess their understanding. Confidential data would remain within approved environments. What matters to me is that students take responsibility for the work they submit and can explain how they arrived at it.